C1 · Parsing–Retrieval Disconnect

Candidate parsers

Qwen3-VL · MinerU · Marker · PaddleOCR

Cleaner-looking ≠ better retrieval

parser	Boundary Clarity	Hit@1
MinerU	0.72 (highest)	**0.20**
Prod (ours)	0.61	**0.55**

parser choice alone moves Hit@1 by **+35.1pp (0.20→0.55; 2.8×)**

appearance metrics measure formatting, not content

(anti-correlation $r = -0.81$, $n = 5$)

C3 · RCPS Selection (+ C2 Diagnose)

C3 · RCPS selection protocol (no training)

held-out Q–A probe, scored over 3 retrievers:

BGE-M3 · *mE5* · *Qwen3-Emb*

extrinsic · retriever-averaged · format-normalised

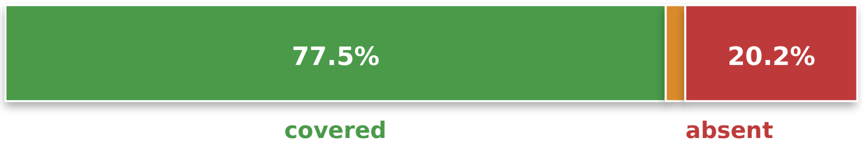
no training · no manual chunk labels

→ rank parsers & chunkers by retrieval

ranks Prod far above MinerU (0.583 vs 0.212)

▼ ***score low? locate which layer is at fault*** ▼

C2 · Coverage diagnostic (no retriever)



absent = **parser-side failure**, set before chunking

no chunker can recover it → fix the parser (C4)

C4 · Bounded Parser Training

C4 · Parser-side training

best-of-K parses from Prod, ranked two ways:

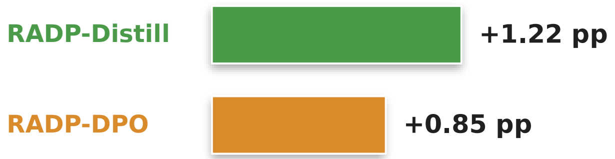
RADP-Distill — edit-distance → ref

vs

RADP-DPO — page-local RCPS

Result

Hit@5 gain over the untuned parser:



substantially overlapping CIs

no evidence retrieval reward beats fidelity control